Australian Mathematics Trust

Australian Mathematics Competition

AN ACTIVITY OF THE AUSTRALIAN MATHEMATICS TRUST

NAME _____

YEAR _____ TEACHER _____

2018
CHINESE VERSION
JUNIOR DIVISION

AUSTRALIAN SCHOOL YEARS 7 and 8

TIME ALLOWED: 75 MINUTES

INSTRUCTIONS AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. NO calculators, maths stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

THE ANSWER SHEET

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

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1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 请问算式 $2+0+1+8$ 之值等于多少?

What is $2+0+1+8$?

(A) 9 (B) 10 (C) 11 (D) 38 (E) 2018

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2. 小佳原本有 \$47，她又得到 \$25 的生日礼金。请问她现在有多少钱?

Callie has \$47 and then gets \$25 for her birthday. How much does she have now?

(A) \$52 (B) \$62 (C) \$65 (D) \$69 (E) \$72

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3. 请问算式 $4\times 10000+3\times 1000+2\times 10+4\times 1$ 之值等于多少?

The value of $4\times 10000+3\times 1000+2\times 10+4\times 1$ is

(A) 4324 (B) 43024 (C) 43204 (D) 430204 (E) 430024

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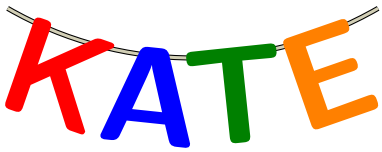
4. 小凯用字母串珠做出右侧这个项链。她将它翻面过来，展现出串珠的背面。请问下列哪一项内的图是现在看起来的样子?

Kate made this necklace from alphabet beads. She put it on the wrong way around, showing the back of the beads. What does this look like?



(A)

(B)

(C)

(D)

(E)

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5. 请问在下午 5：34 之前 58 分钟是什么时刻?

What is the time 58 minutes before 5.34 pm?

(A) 5.32 pm (B) 5.36 pm (C) 6.32 pm (D) 6.12 pm (E) 4.36 pm

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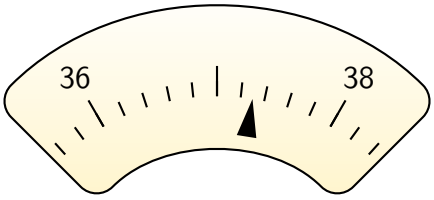
6. 请问这个普遍使用的仪表所指出的值是什么?

What value is indicated on this charisma-meter?



(A) 36.65 (B) 37.65 (C) 38.65 (D) 37.15 (E) 37.3

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7. 从 1000 开始，小丝每次减少 7 向后数。请问小丝数到最小的正整数是什么？

Starting at 1000, Ishrak counted backwards, taking 7 off each time. What was the last positive number he counted?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

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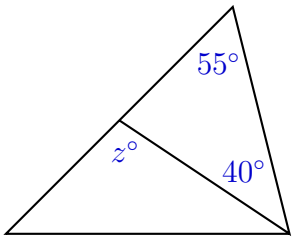
8. 右图中，请问 z 之值是什么？

What is the value of z?

(A) 75 (B) 85 (C) 95 (D) 100 (E) 105



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9. 五位小朋友 A、B、C、D、E 一起玩游戏。他们打算由年龄最大到最小排成一列。

- A 比 B 的年龄大且 B 比 E 的年龄大。
- D 也比 B 的年龄大。
- A 不是年龄最大。
- E 不是年龄最小。

这五位小朋友中，请问谁的年龄第二小？

(A) A (B) B (C) C (D) D (E) E

Five friends (Amelia, Billie, Charlie, David and Emily) are playing together and decide to line up from oldest to youngest.

- Amelia is older than Billie who is older than Emily.
- David is also older than Billie.
- Amelia is not the oldest.
- Emily is not the youngest.

Who is the second-youngest of the five friends?

(A) Amelia (B) Billie (C) Charlie (D) David (E) Emily

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10. 将一条彩带剪成相等的两段。使用掉其中的一段，再使用掉另外一段的三分之一，最后剩下 12 cm 的彩带。请问原来的彩带长度为多少 cm？

A length of ribbon is cut into two equal pieces. After using one piece, one-third of the other piece is used, leaving 12 cm of ribbon. How long, in centimetres, was the ribbon initially?

(A) 24 (B) 32 (C) 36 (D) 48 (E) 50

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 一个数的 1000% 等于 100，请问这个数是什么？

1000% of a number is 100. What is the number?

(A) 0.1 (B) 1 (C) 10 (D) 100 (E) 1000

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12. NORA、ANNE、WARREN 与 ANDREW 四人购买塑料字母以便在生日蛋糕上拼出他们的名字。
他们的生日都在不同的日期，所以他们打算重复使用一些字母在不同的蛋糕上。
请问他们至少共需要多少个塑料字母牌？

Nora, Anne, Warren and Andrew bought plastic capital letters to spell each of their names on their birthday cakes.
Their birthdays are on different dates, so they planned to reuse letters on different cakes.
What is the smallest number of letters they needed?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

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13. 喂养四只狗三天的伙食费为 \$60。若每天每只狗的伙食费都相同，请问喂养七只狗七天的伙食费为多少？

The cost of feeding four dogs for three days is \$60. Using the same food costs per dog per day, what would be the cost of feeding seven dogs for seven days?

(A) \$140 (B) \$200 (C) \$245 (D) \$350 (E) \$420

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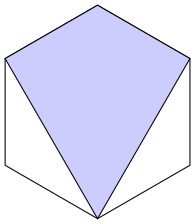
14. 请问这个正六边形的几分之几被涂上阴影？



What fraction of this regular hexagon is shaded?

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{5}$ (E) $\frac{4}{5}$

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15. 小李有一些全等的正方形瓷砖，他将它们以边对边连接成一长条，而构成一个矩形。这个矩形的周长为单独一片瓷砖周长的三倍。请问小李共有多少片瓷砖？

Leila has a number of identical square tiles that she puts together edge to edge in a single row, making a rectangle. The perimeter of this rectangle is three times that of a single tile. How many tiles does she have?

(A) 3 (B) 5 (C) 6 (D) 8 (E) 9

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16. 小杰明年打算选修外语课。他必须选择二门不同的选修课，一门从 A 组选、另一门从 B 组选。

A 组	B 组
华语	华语
日语	德语
西班牙语	阿拉伯语
印度尼西亚语	意大利语

请问小杰有多少种不同的可能选修配对组合?

James is choosing his language electives for next year. He has to choose two different electives, one from Group A and one from Group B.

Group A	Group B
Mandarin	Mandarin
Japanese	German
Spanish	Arabic
Indonesian	Italian

How many different pairs of elective combinations are possible?

(A) 7 (B) 8 (C) 12 (D) 15 (E) 16

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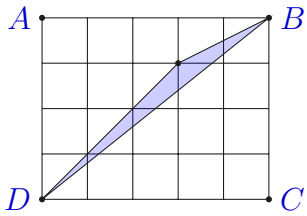
17. 在右图中，$ABCD$ 是一个 $5\,\text{cm} \times 4\,\text{cm}$ 的矩形。每个小方格是 $1\,\text{cm} \times 1\,\text{cm}$。请问阴影部分的面积为多少 cm^2?



In the diagram, $ABCD$ is a $5\,\text{cm} \times 4\,\text{cm}$ rectangle and the grid has $1\,\text{cm} \times 1\,\text{cm}$ squares. What is the shaded area, in square centimetres?

(A) 1 (B) 1.5 (C) 0.5 (D) 2 (E) 3

---

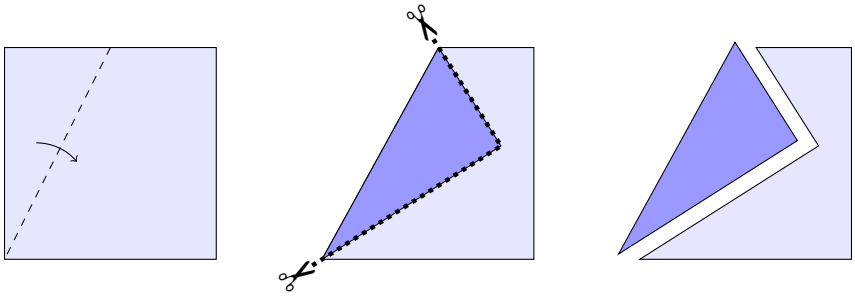
18. 将一些数填入右侧的方格表内，使得每一列、每一行与每条对角线上的数之和都等于 18。
请问在四个角落上的数之总和等于多少?

	6	
	4	

Fill in this diagram so that each of the rows, columns and diagonals adds to 18.
What is the sum of all the corner numbers?

(A) 20 (B) 22 (C) 23 (D) 24 (E) 25

19. 将一张正方形纸片沿着顶边的中点与左下方的顶点之连线向内折迭。再将下层的纸片沿着上层纸片的边缘剪开，如下图所示。



将折迭的这片展开，请问下列哪一项内对于这两片纸片的描述是正确的？

(A) 一片为筝形、一片为五边形且它们的面积相等
(B) 一片为矩形、一片为五边形且它们的面积相等
(C) 一片为等腰三角形、一片为五边形且五边形的面积较大
(D) 一片为筝形、一片为五边形且筝形的面积较小
(E) 一片为矩形、一片为五边形且矩形的面积较大

A square of paper is folded along a line that joins the midpoint of one side to a corner. The bottom layer of paper is then cut along the edges of the top layer as shown.

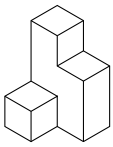
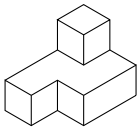
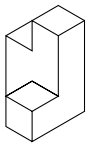
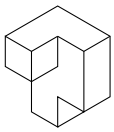
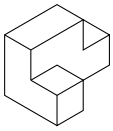
When the folded piece is unfolded, which of the following describes all the pieces of paper?

(A) a kite and a pentagon of equal area
(B) a rectangle and a pentagon of equal area
(C) an isosceles triangle and a pentagon, with the pentagon of larger area
(D) a kite and a pentagon, with the kite smaller in area
(E) a rectangle and a pentagon, with the rectangle larger in area

---

20. 一个 3 维的积木是由六个全等的正立方体黏贴在一起而构成的。以下各项内的积木中，有四个是从不同角度观看此积木的图形，而其中有一个是不同的积木。请问哪一项内的图是这个不同的积木？

A 3-dimensional object is formed by gluing six identical cubes together. Four of the diagrams below show this object viewed from different angles, but one diagram shows a different object. Which diagram shows the different object?

(A) 
(B) 
(C) 
(D) 
(E) 

21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 我们定义一个月的千分之一为*一毫月*。请问一毫月大约等于多久?

(A) 20 秒　(B) 70 秒　(C) 8 分钟　(D) 40 分钟　(E) 3 小时

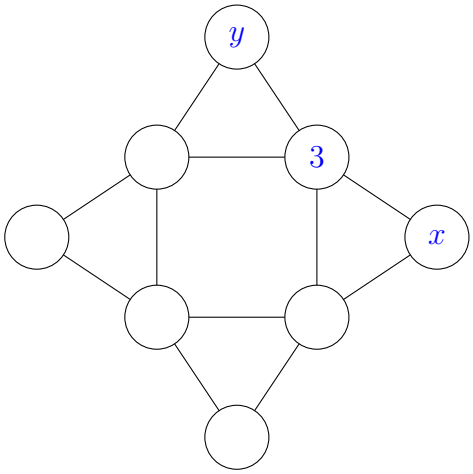
Approximately how long is a *millimonth*, defined to be one-thousandth of a month?

(A) 20 seconds　(B) 70 seconds　(C) 8 minutes　(D) 40 minutes　(E) 3 hours

---

22. 将从 1 到 8 的数不重复地填入右图所示的圆圈内，其中数 3 已经填入，如图所示。
每一个三角形上的三个数之和都相等。
在中央正方形角落的四个数之和为 20。
请问 $x+y$ 等于多少?

The numbers from 1 to 8 are entered into the eight circles in this diagram, with the number 3 placed as shown.
In each triangle, the sum of the three numbers is the same.
The sum of the four numbers which are at the corners of the central square is 20.
What is $x+y$?



(A) 10　(B) 11　(C) 12　(D) 13　(E) 14

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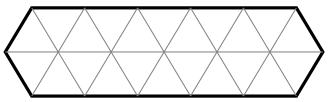
23. 一个长且窄的六边形是由 22 个边长为 1 单位的正三角形组成的。
欲用 11 个边长为 1 单位的菱形 (它是由二个边长为 1 单位的正三角形组成的) 将这个六边形盖满，请问共有多少种不同的方法?

六边形
Hexagon



菱形
Rhombus

A long narrow hexagon is composed of 22 equilateral triangles of unit side length.
In how many ways can this hexagon be tiled by 11 rhombuses of unit side length?

(A) 6　(B) 8　(C) 9　(D) 12　(E) 16

---

24. 在以下的小方格内填入加号或减号使得其结果为最小可能的正数:

$$\square\,\frac{1}{3}\,\square\,\frac{1}{4}\,\square\,\frac{1}{5}\,\square\,\frac{1}{6}\,\square\,\frac{1}{7}$$

请问此结果满足下列哪一项?

(A) 介于 0 与 $\frac{1}{100}$ 之间　(B) 介于 $\frac{1}{100}$ 与 $\frac{1}{50}$ 之间　(C) 介于 $\frac{1}{50}$ 与 $\frac{1}{20}$ 之间　(D) 介于 $\frac{1}{20}$ 与 $\frac{1}{10}$ 之间　(E) 介于 $\frac{1}{10}$ 与 1 之间

In this expression

$$\square\,\frac{1}{3}\,\square\,\frac{1}{4}\,\square\,\frac{1}{5}\,\square\,\frac{1}{6}\,\square\,\frac{1}{7}$$

we place either a plus sign or a minus sign in each box so that the result is the smallest positive number possible. The result is

(A) between 0 and $\frac{1}{100}$　(B) between $\frac{1}{100}$ and $\frac{1}{50}$　(C) between $\frac{1}{50}$ and $\frac{1}{20}$　(D) between $\frac{1}{20}$ and $\frac{1}{10}$　(E) between $\frac{1}{10}$ and 1

---

25. 在以下减式中，第一个数有 100 个数码、第二个数有 50 个数码。

$$\underbrace{111\ldots\ldots111}_{100\text{ 位数}} - \underbrace{222\ldots222}_{50\text{ 位数}}$$

请问所得结果的数码和是多少?

In this subtraction, the first number has 100 digits and the second number has 50 digits.

$$\underbrace{111\ldots\ldots111}_{100\text{ digits}} - \underbrace{222\ldots222}_{50\text{ digits}}$$

What is the sum of the digits in the result?

(A) 375　(B) 420　(C) 429　(D) 450　(E) 475

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 只使用数码 0、1、2，右图这个正立方体每个面上的数都互不相同。任意在相对面上的两个数之和都等于一个相同的三位数。
请问这两个数之和的最大可能值是什么？

Using only digits 0, 1 and 2, this cube has a different number on each face.
Numbers on each pair of opposite faces add to the same 3-digit total.
What is the largest that this total could be?

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27. 有一个三位数减掉它的数码和后等于这个三位数的数码和之平方。请问这个三位数是什么？

I have a three-digit number, and I add its digits to create its *digit sum*. When the digit sum of my number is subtracted from my number, the result is the square of the digit sum. What is my three-digit number?

---

28. 从甲地到乙地的铁路长度为 999 km。在沿路上每 km 都有一个如下图所示的路标指示与两地之距离。

$$\boxed{0|999} \quad \boxed{1|998} \quad \boxed{2|997} \quad \boxed{3|996} \quad \cdots \quad \boxed{998|1} \quad \boxed{999|0}$$

请问总共有多少个路标上只恰使用二种不同的数码?

A road from Tamworth to Broken Hill is 999 km long. There are road signs each kilometre along the road that show the distances (in kilometres) to both towns as shown in the diagram.

How many road signs are there that use exactly two different digits?

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29. 在以下的乘式中，X、Y 与 Z 为非零的相异数码。

$$\begin{array}{cccc} & X & Y & Z \\ \times & & 1 & 8 \\ \hline Z & X & Y & Y \end{array}$$

请问三位数 $\overline{XYZ}$ 是什么?

In the multiplication shown, X, Y and Z are different non-zero digits.

What is the three-digit number XYZ?

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30. 令 A 为一个可被 9 整除的 2018 位数，令 B 为 A 的所有数码相加、C 为 B 的所有数码和。请问 C 的所有可能值之总和是什么?

Let A be a 2018-digit number which is divisible by 9. Let B be the sum of all digits of A and C be the sum of all digits of B. Find the sum of all possible values of C.

Answer Key

QUES	MP	UP	J	I	S
1	C	D	C	D	D
2	E	E	E	E	C
3	A	D	B	D	D
4	B	A	A	D	B
5	D	E	E	E	A
6	D	D	E	B	A
7	A	D	E	C	C
8	E	C	C	B	C
9	D	B	E	A	E
10	A	B	C	B	A
11	D	C	C	C	D
12	B	A	B	A	E
13	E	A	C	D	E
14	B	C	B	E	E
15	A	E	B	A	C
16	C	C	D	E	B
17	B	C	B	E	B
18	A	B	D	C	A
19	C	B	A	E	D
20	C	A	A	A	C
21	D	A	D	A	C
22	B	E	D	D	A
23	D	E	A	C	C
24	E	D	C	D	B
25	D	D	D	B	E
26	16	918	321	156	90
27	561	321	156	40	97
28	936	509	40	435	90
29	918	252	529	505	118
30	509	693	90	118	101